Compiler Design

No	Problem Name
01	Write a C program for developing a lexical analyzer (LA) that will eliminate white spaces form a source program in C and collect numbers.
02	Write a C program for developing a lexical analyzer (LA) that will eliminate white spaces form a source program in C and collect numbers as token and then also display the token value as attribute.
03	Write a C program for developing a lexical analyzer (LA) that will recognize the variables in a source program.
04	Write a C program for developing a lexical analyzer (LA) that will recognize the keywords in a source program.
05	Design a compiler front-end based on syntax-directed translation technique that will function as an infix translator for a language consists of sequence of expressions terminated by semicolon.
06	Write a program to remove left factoring.
07	Write a program to check whether a string belongs to the grammar or not.
08	Write a C program to implement Shift-Reduce Parser.
09	Write a C program to implement a recursive descent parser.

1. Write a C program for developing a lexical analyzer (LA) that will eliminate white spaces form a source program in C and collect numbers.

```
#include<iostream>
#include<stdio.h>
using namespace std;
int sign_check(char pre_value){
    if(pre_value=='+'||pre_value=='-'||pre_value=='*'||pre_value=='/'||pre_value=='%'||pre_value=='=')
        return 1;
    else
        return 0;
}
int main()
{
    int n,len,s,i;
    char pre_value=' ';
    string str;
    FILE *f2;
    freopen("1 Input.txt","r",stdin);
    f2=fopen("1 Output.txt","w");
    printf("Collect no. given below\n");
    while(getline(cin,str)){
        len =str.size();
        for(i=0;i<len;i++){
            if(str[i]!=' ')
                putc(str[i],f2);
        }
        putc('\n',f2);
        for(i=0;i<len;i++){
            if(str[i]>='0'&&str[i]<='9'){
                s=str[i]-'0';
```

```
            for(i=i+1;i<len;i++){
                if(str[i]>='0'&&str[i]<='9')
                    s=s*10+(str[i]-'0');
                else
                    break;
            }
            i--;
            if(pre_value=='['&&str[i+1]==']')
                cout<<s<<endl;
            else if((sign_check(pre_value)||pre_value==' ')&&(sign_check(str[i+1])||str[i+1]==';'||i==len1))
                cout<<s<<endl;
        }
        else
            pre_value=str[i];
      }
   }
   return 0;
}
```

Input:

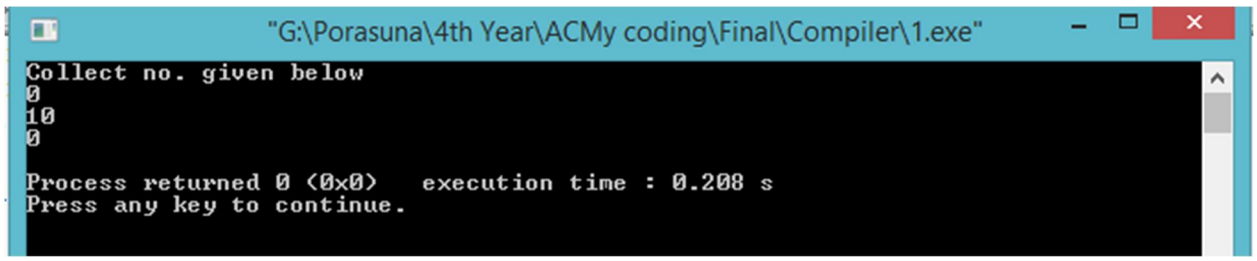
```
#include<stdio.h>
#include<iostream>
using namespace std;
int main()
{
        char t,f;
        int n;
        FILE *f1,*f2;
        f1=fopen("basir.txt","r");
        f2=fopen("rajib.txt","w");
        while(t=getc(f1)!=EOF){
```

```
            if(t!=''&&isdigit(t))
                putc(t,f2);
            else{
                n=0;
                while(isdigit(t)){
                    putc(t,f2);
                    n=n*10+(t-'0');
                    t=getc(f1);
                }
                if(n){
                    putc(t,f2);
                    cout<<n<<endl;
                }
            }
        }
        return 0;
}
```

Output:

```
#include<stdio.h>
#include<iostream>
usingnamespacestd;
intmain()
{
        chart,f;
        intn;
        FILE*f1,*f2;
        f1=fopen("basir.txt","r");
        f2=fopen("rajib.txt","w");
        while(t=getc(f1)!=EOF){
```

```
            if(t!="&&isdigit(t))
                putc(t,f2);
            else{
                n=0;
                while(isdigit(t)){
                    putc(t,f2);
                    n=n*10+(t-'0');
                    t=getc(f1);
                }
                if(n){
                    putc(t,f2);
                    cout<<n<<endl;
                }
            }
        }
        return0;
}
```



2. Write a C program for developing a lexical analyzer (LA) that will eliminate white spaces form a source program in C and collect numbers as token and then also display the token value as attribute.

```
#include<iostream>
#include<stdio.h>
using namespace std;
int punchuation_check(char pre_value){
    if(pre_value=='+'||pre_value=='-'||pre_value=='*'||pre_value=='/'||pre_value=='%')
        return 1;
    else if(pre_value=='['||pre_value==']'||pre_value=='{'||pre_value=='}'||pre_value=='('||pre_value==')')
        return 1;
    else if(pre_value=='"'||pre_value==';'||pre_value==':'||pre_value==','||pre_value=='.'||pre_value=='\')
        return 1;
    else
        return 0;
}
int keyword(string str){
if(str=="int"||str=="char"||str=="float"||str=="double"||str=="string"||str=="typedef"||str=="long"||str=="str
ing")
        return 1;
    else if(str=="while"||str=="do"||str=="for"||str=="if"||str=="else"||str=="switch"
||str=="case"||str=="break"||str=="continue")
        return 1;
    else if(str=="const"||str=="goto"||str=="static"||str=="union"||str=="return"||str=="default"
||str=="short"||str=="signed"||str=="unsigned"||str=="void")
        return 1;
    else
        return 0;
}
int sign_check(char pre_value){
    if(pre_value=='+'||pre_value=='-'||pre_value=='*'||pre_value=='/'||pre_value=='=')
```

```
        return 1;
    else
        return 0;
}
int main()
{
    char t,f;
    int n,checks,s,len;
    string str,stt;
    FILE *f2;
    freopen("2 Input.txt","r",stdin);
    char pre_value='1';
    cout<<"token    token value as attributes\n";
    cout<<"==================================\n";
    while(getline(cin,str)){
        pre_value='1';
        stt="";
        len=str.size();
        for(int i=0;i<len;i++){
            if(punchuation_check(str[i])){
                cout<<"Punchuation "<<str[i]<<"\n";
                pre_value=str[i];
            }
           else if(str[i]=='<' or str[i]=='>' or str[i]=='='){
                if(i+1<len){
                    if(str[i+1]=='='){
                        cout<<"CO          "<<str[i]<<str[i+1]<<endl;
                        i++;
                    }
                    else
```

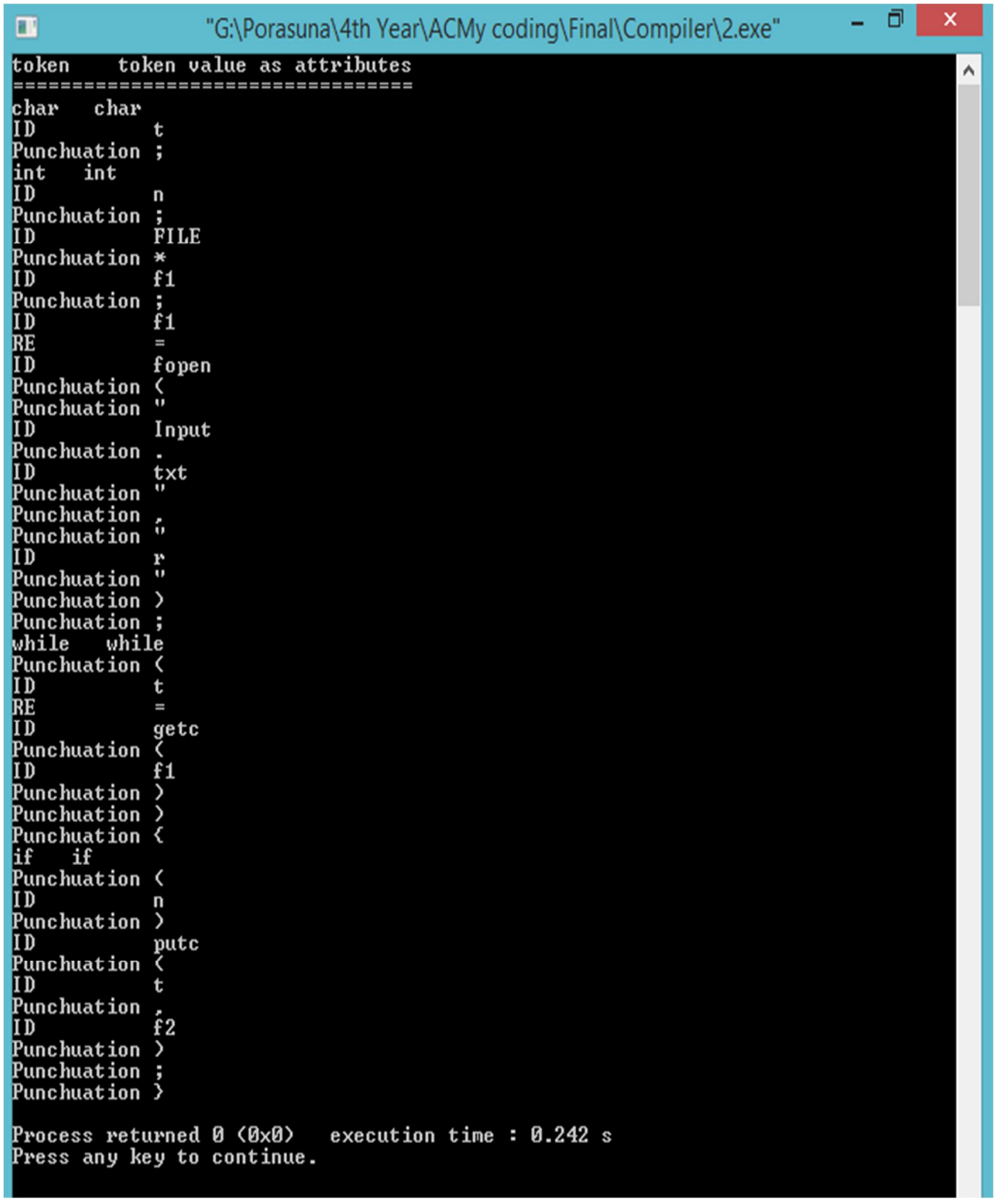
```
            cout<<"RO          "<<str[i]<<endl;
        }
        pre_value=str[i];
    }
    else if(str[i]==' ')
        pre_value=str[i];
    else if((str[i]>='0' and str[i]<='9') and sign_check(pre_value)){
        s=0;
        s=s*10+str[i]-'0';
        for(i=i+1;i<len;i++){
            if((str[i]>='0' and str[i]<='9'))
                s=s*10+str[i]-'0';
            else
                break;
        }
        i--;
        if(pre_value=='[' and str[i+1]==']')
            cout<<"num         "<<s<<"\n";
        else if( sign_check(pre_value) and  i==len-1 )
            cout<<"num         "<<s<<"\n";
        else if( sign_check(pre_value) and (sign_check(str[i+1]) or str[i+1]==';'))
            cout<<"num         "<<s<<"\n";
    }
    else{
        stt+=str[i];
        if(punchuation_check(str[i+1]) or str[i+1]==' ' or str[i+1]=='=' or str[i+1]=='<'){
            if(keyword (stt))
                cout<<"Keyword     "<<stt<<"\n";
            else
                cout<<"ID          "<<stt<<"\n";
```

```
                stt="";
            }
            pre_value=str[i];
        }
    }
  }
  return 0;
}
```

Input:

```
char t;
int n;
FILE *f1;
f1=fopen("Input.txt","r");
while(t=getc(f1)){
  if(n)
    putc(t,f2);
}
```

Output:


"G:\Porasuna\4th Year\ACMy coding\Final\Compiler\2.exe"
token token value as attributes
==========
char char
ID t
Punchuation ;
int int
ID n
Punchuation ;
ID FILE
Punchuation *
ID f1
Punchuation ;
ID f1
RE =
ID fopen
Punchuation (
Punchuation "
ID Input
Punchuation .
ID txt
Punchuation "
Punchuation ,
Punchuation "
ID r
Punchuation "
Punchuation)
Punchuation ;
while while
Punchuation (
ID t
RE =
ID getc
Punchuation (
ID f1
Punchuation)
Punchuation)
Punchuation {
if if
Punchuation (
ID n
Punchuation)
ID putc
Punchuation (
ID t
Punchuation ,
ID f2
Punchuation)
Punchuation ;
Punchuation }
Process returned 0 (0x0) execution time : 0.242 s
Press any key to continue.

3. Write a C program for developing a lexical analyzer (LA) that will recognize the variables in a source program.

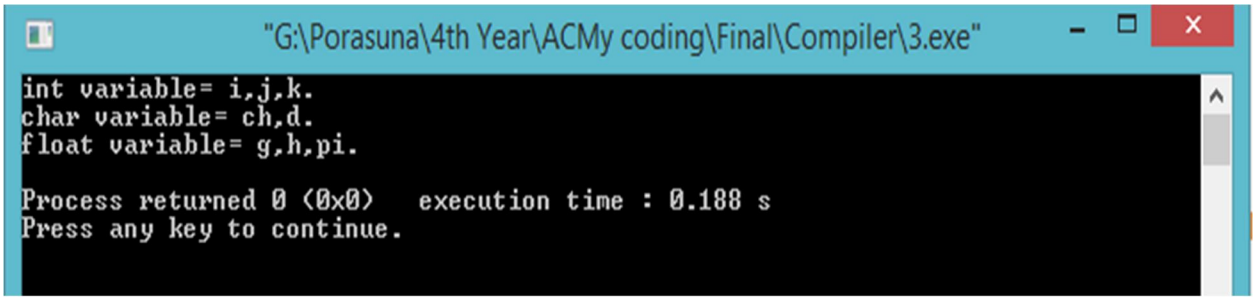
```
#include<stdio.h>
#include<iostream>
using namespace std;
int check(string str){
   if(str=="int"||str=="char"||str=="float"||str=="double"||str=="long"||str=="long long"||str=="signed"||str=="unsigned")
      return 1;
   else
      return 0;
}
int main()
{
   freopen("3 Input.txt","r",stdin);
   string str;

   while(cin>>str){
      if(check(str)){
         cout<<str<<" variable= ";
         int k=0;
         while(1){
            char ch=getchar();
            if(ch==';')
               break;
            if(k==0&&ch!=' ')
               cout<<ch;
            if(ch=='=')
               k=1;
            if(ch==',')
```

```
            k=0;


        }
        cout<<"."<<endl;
      }
   }
   return 0;
}
```

Input:

```
void main()
{
        int i,j,k;
        char ch,d;
        float g,h,pi;
        for(i=0;i<5;i++)
                printf("%d ",i);
        return 0;
}
```

Output:



4. Write a C program for developing a lexical analyzer (LA) that will recognize the keywords in a source program.

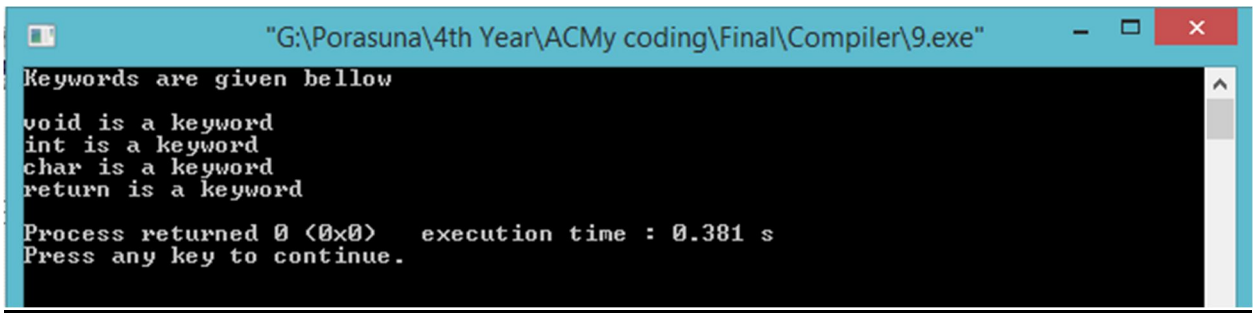
```
#include<bits/stdc++.h>

using namespace std;

int check(string s){


if(s=="auto"||s=="int"||s=="char"||s=="long"||s=="float"||s=="double"||s=="struct"||s=="if"||s=="else"||s==
"break"||s=="continue"||


s=="while"||s=="do"||s=="for"||s=="return"||s=="signed"||s=="unsigned"||s=="default"||s=="goto"||s=="ca
se"||s=="sizeof"||s=="short"||

      s=="switch"||s=="void"||s=="static"||s=="typdef")

      return 1;

   else

       return 0;

}

int main()

{

   char t,f,ch;

   int n,k;

   int anu;

   string st;

   freopen("9 Input.txt","r",stdin);

   char pre_value='1';

   printf("Keywords are given bellow\n\n");


   while(cin>>st){

      if(check(st)){

         cout<<st<<" is a keyword\n";

      }

   }
```

```
    return 0;
}
```

Input:

```
#include<iostream>
using namespace std;


void main()
{
        int i,a,b,c;
        char ch;
        string str;
        for(i=1;i<=15;i++)
                cout<<"The number is: "<<i<<endl;
        return 0;
}
```

Output:



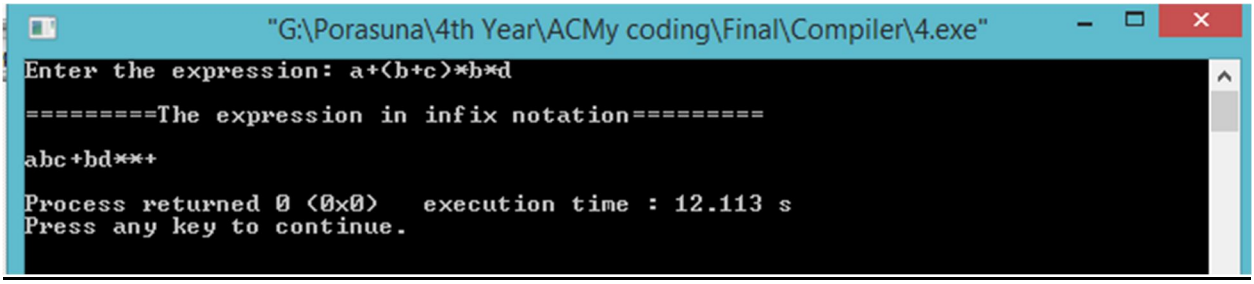
5. Design a compiler front-end based on syntax-directed translation technique that will function as an infix translator for a language consists of sequence of expressions terminated by semicolon.

```
#include<bits/stdc++.h>
using namespace std;
int len,i;
char ch;

void infix(string str){
    stack<char> stack_string;
    str='('+str+')';
    len=str.size();

    for(i=0;i<len;i++){
        if(str[i]>='a'&& str[i]<='z')
            cout<<str[i];
        else if(str[i]=='(')
            stack_string.push('(');
        else if(str[i]==')'){
            while(stack_string.top()!='('){
                cout<<stack_string.top();
                stack_string.pop();
            }
            stack_string.pop();
        }
        else if(str[i]=='+'||str[i]=='-'){
            while(1){
                ch=stack_string.top();
                if(ch=='+'||ch=='-')
                    break;
```

```
            else if(ch=='(')
                break;
            cout<<ch;
            stack_string.pop();
        }
        stack_string.push(str[i]);
    }
    else if(str[i]=='*'||str[i]=='/'){
        while(1){
            ch=stack_string.top();
            if(ch=='+'||ch=='-'||ch=='*'||ch=='/')
                break;
            else if(ch=='(')
                break;
            cout<<ch;
            stack_string.pop();
        }
        stack_string.push(str[i]);
    }
    else if(str[i]=='^')
        stack_string.push(str[i]);
  }
}

int main()
{
   string str;
   cout<<"Enter the expression: ";
   cin>>str;
   cout<<"\n=========The expression in infix notation=========\n\n";
```

```
    infix(str);
    cout<<endl;
    return 0;
}
```

Input & Output:



6. Write a program to remove left factoring.

```
#include<bits/stdc++.h>
using namespace std;

int main()
{
    int len,i,j,k,kk,check=0;
    string str;
    while(1){
        cout<<"Enter the string: ";
        cin>>str;
        len=str.size();

        for(i=0;i<len;i++){
            if(str[i]=='|')
                break;
        }
```

```
j=i;
kk=j+1;

for(k=0;k<len;k++,kk++){
   if(str[k]!=str[kk]){
      check=1;
      break;
   }
}

if(check && k!=0){
   cout<<"A-> ";
   for(i=0;i<k;i++)
      cout<<str[i];
   cout<<"A'"<<endl;
   cout<<"A'-> ";

   if(k==j)
      cout<<"#";

   for(i=k;i<j;i++)
      cout<<str[i];
   cout<<"|";

   for(i=j+k+1;i<len;i++)
      cout<<str[i];
   cout<<endl;
}else{
   cout<<"A-> ";
   for(i=0;i<j+1;i++)
```

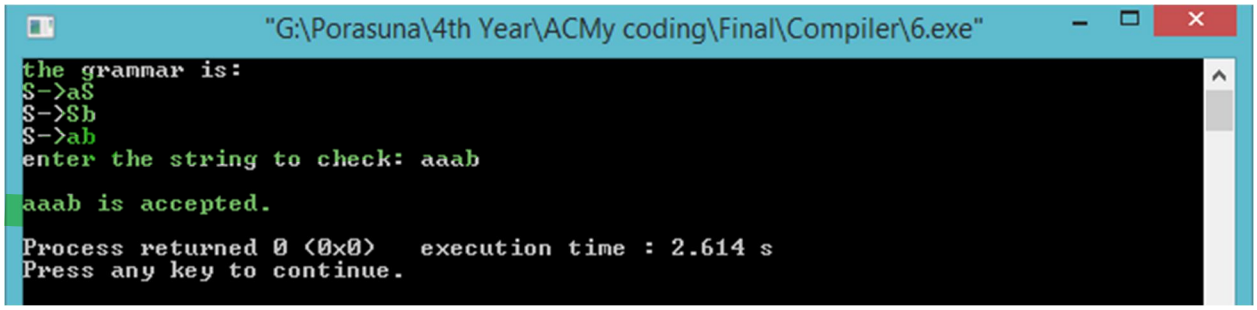
```
            cout<<str[i];
        cout<<"A'"<<endl;
        cout<<"A'-> ";

        for(i=j+1;i<len;i++)
            cout<<str[i];
        cout<<endl;
      }
      cout<<endl;
   }
   return 0;
}
```

Input & Output:

7. Write a program to check whether a string belongs to the grammar or not.

```
#include<bits/stdc++.h>
using namespace std;
int main()
{
    string str;
    cout<<"the grammar is: \nS->aS \nS->Sb \nS->ab\n";
    cout<<"enter the string to check: ";
    cin>>str;
    int len=str.size();
    int flag=0,i,a,b;
    for(i=0;i<len;i++){
        if(str[i]!='a'){
            a=i-1;
            break;
        }
    }
    for(i=len-1;i>=0;i--){
        if(str[i]!='b'){
            b=i+1;
            break;
        }
    }
    if(a>=0 && b<len && a+1==b)
        flag=1;
    if(flag==1)
        cout<<endl<<str<<" is accepted."<<endl;
    else
        cout<<endl<<str<<" is not accepted."<<endl;
    return 0;}
```

Input & Output:



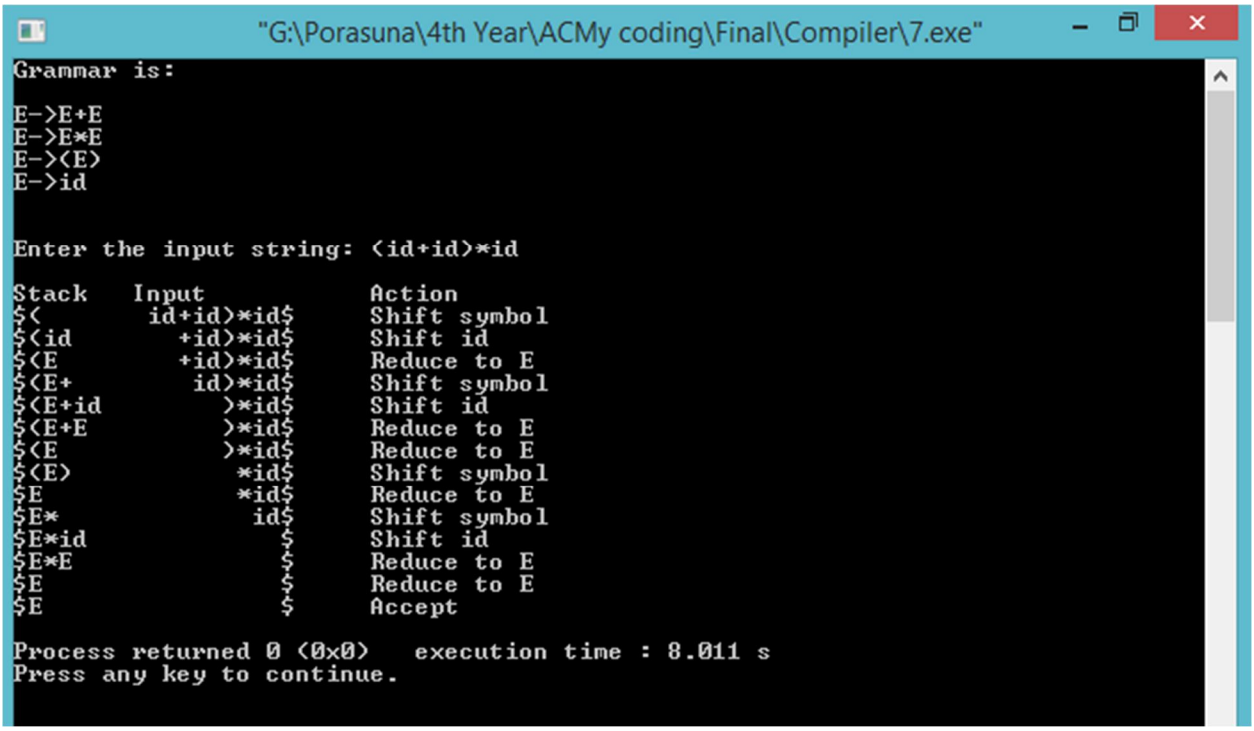
8. Write a C program to implement Shift-Reduce Parser.

```
//(id+id)*id
#include<bits/stdc++.h>
#include<string.h>
using namespace std;

string str,shift_action="Shift ",reduce_action="Reduce to E ";
char stuck[100];
int i,j,len,k;

void checking(){
    for(k=0;k<len;k++){
      if(stuck[k]=='i'&&stuck[k+1]=='d'){
         stuck[k]='E';
         stuck[k+1]='\0';
         cout<<"$"<<stuck<<"\t"<<str<<"$\t"<<reduce_action<<endl;
         j++;
      }
    }
    for(k=0;k<len;k++){
      if(stuck[k]=='E' && stuck[k+1]=='+' && stuck[k+2]=='E'){
         stuck[k]='E';
```

```
            stuck[k+1]='\0';
            stuck[k+2]='\0';
            cout<<"$"<<stuck<<"\t"<<str<<"$\t"<<reduce_action<<endl;
            i=i-2;
        }
    }
    for(k=0;k<len;k++){
        if(stuck[k]=='E'&&stuck[k+1]=='*'&&stuck[k+2]=='E'){
            stuck[k]='E';
            stuck[k+1]='\0';
            stuck[k+1]='\0';
            cout<<"$"<<stuck<<"\t"<<str<<"$\t"<<reduce_action<<endl;
            i=i-2;
        }
    }
    for(k=0;k<len;k++){
        if(stuck[k]=='('&&stuck[k+1]=='E'&&stuck[k+2]==')'){
            stuck[k]='E';
            stuck[k+1]='\0';
            stuck[k+1]='\0';
            cout<<"$"<<stuck<<"\t"<<str<<"$\t"<<reduce_action<<endl;
            i=i-2;
        }
    }
}
int main()
{
    puts("Grammar is:\n\nE->E+E\nE->E*E\nE->(E)\nE->id\n\n");
    cout<<"Enter the input string: ";
    cin>>str;
```

```
    len=str.size();
    puts("\nStack\tInput\t\tAction");
    //cout<<"$\t"<<str<<"$"<<"\t\t"<<shift_action;

    for(i=0,j=0;j<len;i++,j++){
        if(str[j]=='i'&&str[j+1]=='d'){
            stuck[i]=str[j];
            stuck[i+1]=str[j+1];
            stuck[i+2]='\0';
            str[j]=' ';
            str[j+1]=' ';
            cout<<"$"<<stuck<<"\t"<<str<<"$"<<"\t"<<shift_action<<"id"<<endl;
            checking();
        }
        else{
            stuck[i]=str[j];
            stuck[i+1]='\0';
            str[j]=' ';
            cout<<"$"<<stuck<<"\t"<<str<<"$"<<"\t"<<shift_action<<"symbol"<<endl;
            checking();
        }
    }
    if(str[len-1]==' ')
        cout<<"$"<<stuck<<"\t"<<str<<"$\t"<<"Accept"<<endl;

    return 0;
}
```

Input & Output:



9. Write a C program to implement a recursive descent parser.

//E->TE'

//E'->+TE'|NULL

//T->FT'

//T'->*F'|NULL

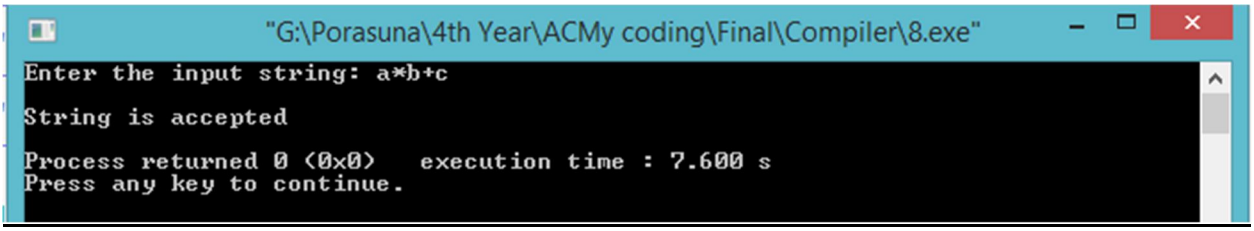
//F->(E)|a|b|c

```
#include<bits/stdc++.h>
using namespace std;
int i,error;
string str;

void E();
void Eds();
void T();
```

```
void Tds();
void F();

int main()
{
    cout<<"Enter the input string: ";
    cin>>str;
    E();
    if(i==str.size()&&error==0)
        cout<<endl<<"String is accepted"<<endl;
    else
        cout<<endl<<"String is rejected"<<endl;
    return 0;
}

void E(){
    T();
    Eds();
}

void Eds(){
    if(str[i]=='+'){
        i++;
        T();
        Eds();
    }
}

void T(){
    F();
```

```
    Tds();
}

void Tds(){
    if(str[i]=='*'){
        i++;
        F();
        Tds();
    }
}

void F(){
    if(str[i]=='a'||str[i]=='b'||str[i]=='c')
        i++;
    else if(str[i]='('){
        E();
        if(str[i]==')')
            i++;
        else
            error=1;
    }
    else
        error=1;
}
```

Input & Output:



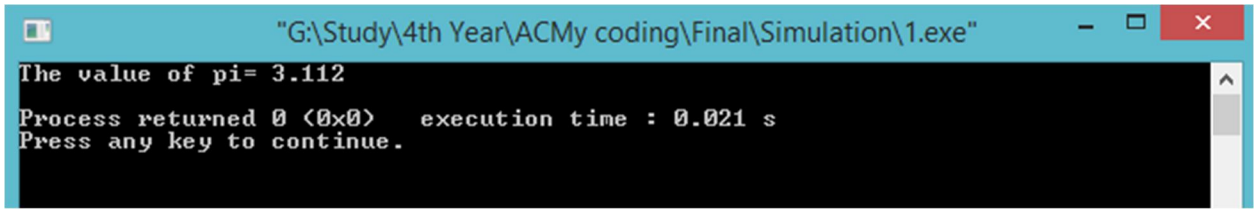
System Simulation

No	Problem Name
01	Develop a computer program for determining the value of π and run the simulation for 1000 observations.
02	Write a C Program to simulate pure pursuit problem.
03	Write a C program to generate the area of an irregular figure using monte-carlo method.
04	Write a C program to determine the value of numerical integration using monte-carlo method.
05	Write a C program to determine the value of π by monte-carlo method.
06	Write a C program for Mid-square Method of Generating 4-digit Random Numbers.
07	Write a C program to generate variables from NEGATIVE BINOMIAL distribution having Parameters k (number of successes) and p (probability of success in trial).
08	Write a C program to generate random variable from a Normal distribution having mean mue and standard deviation sigma.
09	

1. Develop a computer program for determining the value of π and run the simulation for 1000 observations.

```
#include<bits/stdc++.h>
#include<math.h>
using namespace std;
int main()
{
   float r1,r2,m=0;
   int i,n=1000;
   for(i=1;i<=n;i++){
      r1=rand()/32768.0;
      r2=rand()/32768.0;
      if(r2<=sqrt(1-r1*r1))
         m++;
   }
   cout<<"The value of pi= "<<(m*4)/n<<endl;
   return 0;
}
```

Input & Output:

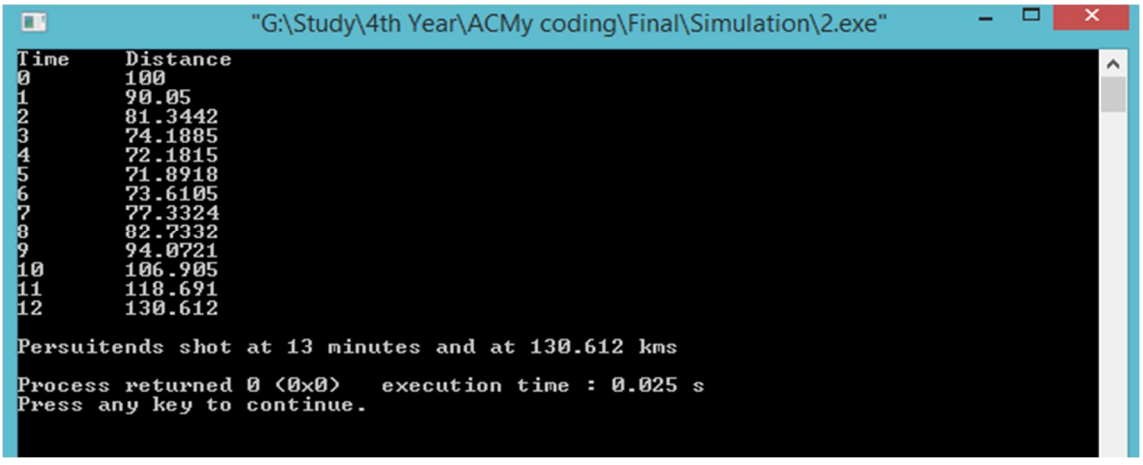


2. Write a C Program to simulate pure pursuit problem.

```
#include<bits/stdc++.h>
using namespace std;
int main()
{
   float xb[]={100,110,120,129,140,149,158,168,179,188,198,209,219,226,234,240};
```

```
    float yb[]={0,3,6,10,15,20,26,32,37,34,30,27,23,19,16,14};
    float dist,time_limit=12,xf[20],yf[20],s=20;
    int i;
    xf[0]=0;
    yf[0]=0;
    cout<<"Time\tDistance\n";
    for(i=0;i<=12;i++){
        dist=sqrt(pow(yf[i]-yb[i],2)+pow(xf[i]-xb[i],2));
        cout<<i<<"\t"<<dist<<endl;
        if(i>time_limit)
            printf("\nTarget escapes");
        if(dist<=10)
            break;
        else{
            xf[i+1]=xf[i]+s*(xb[i]-xf[i])/dist;
            yf[i+1]=xf[i]+s*(yb[i]-yf[i])/dist;
        }
    }
    cout<<"\nPersuitends shot at "<<i<<" minutes and at "<<dist<<" kms\n";
    return 0;
}
```

Input & Output:

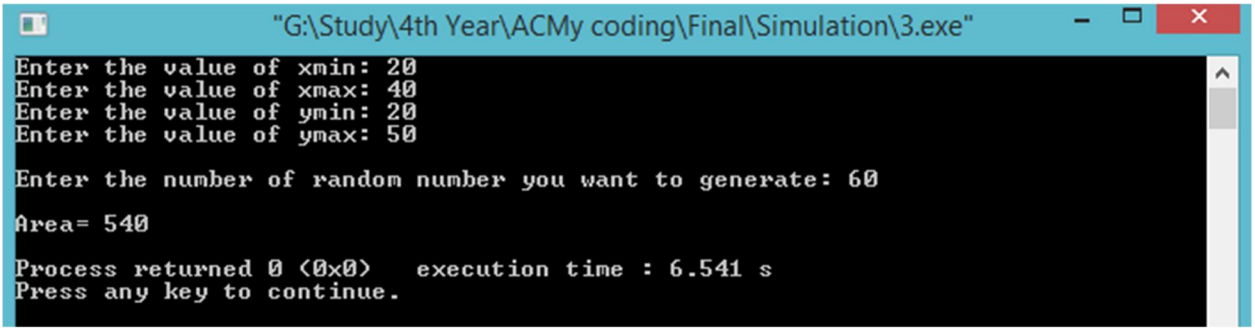


3. Write a C program to generate the area of an irregular figure using monte-carlo method.

```
#include<bits/stdc++.h>
using namespace std;
int main()
{
   int i,xmin,xmax,ymin,ymax,n;
   float area,mf,r1,r2,m,a,x,y;
   cout<<"Enter the value of xmin: ";
   cin>>xmin;
   cout<<"Enter the value of xmax: ";
   cin>>xmax;
   cout<<"Enter the value of ymin: ";
   cin>>ymin;
   cout<<"Enter the value of ymax: ";
   cin>>ymax;
   cout<<"\nEnter the number of random number you want to generate: ";
   cin>>n;
   if(xmin<=0)
      xmax=xmax-xmin;
   if(ymin<=0)
      ymax=ymax-ymin;
   if(ymax>xmin)
      mf=ymax+10;
   else
      mf=xmax+10;
   for(i=1;i<=n;i++){
      x=rand()/32768.0;
      y=rand()/32768.0;
      x*=mf;
      y*=mf;
```

```
        if(x>=xmin&&x<=xmax&&y>=ymin&&y<=ymax)
            m++;
    }
    mf*=mf;
    area=m/float(n);
    area*=mf;
    cout<<"\nArea= "<<area<<endl;
    return 0;
}
```

Input & Output:

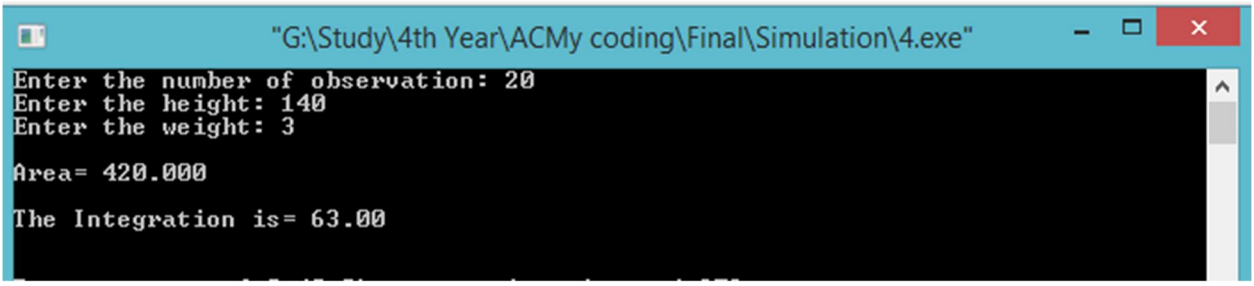


4. Write a C program to determine the value of numerical integration using monte-carlo method.

```
#include<stdio.h>
#include<stdlib.h>
#include<math.h>
int n,i=1;
float r,x,y,x3,h,w,area,I,m,u;
int main()
{
    printf("Enter the number of observation: ");
    scanf("%d",&n);
    printf("Enter the height: ");
```

```
    scanf("%f",&h);
    printf("Enter the weight: ");
    scanf("%f",&w);
    area=h*w;
    printf("\nArea= %.3f\n",area);
    do{
       r=rand()/32768.0;
       r*=50;
       if(r>20&&r<50){
          x=0.1*r;
          y=rand()*140/32768.0;
          x3=x*x*x;
          if(y<=x3)
             m++;
          i++;
       }
    }while(i!=n);
    u=n;
    I=(m/u)*area;
    printf("\nThe Integration is= %.2f\n\n",I);
    return 0;
}
```

Input & Output:

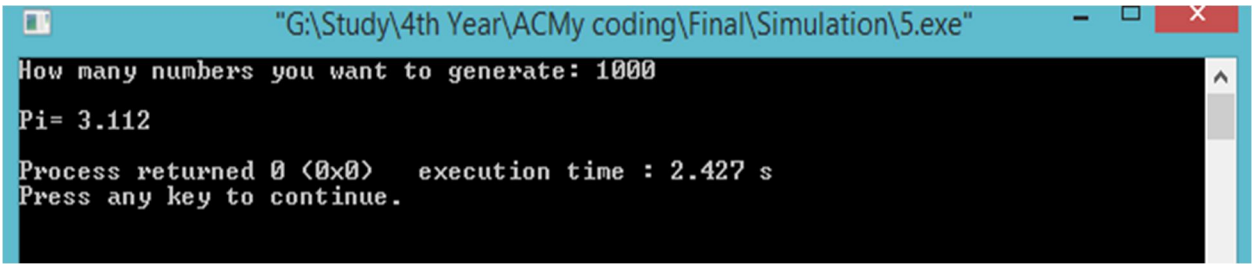


N.B: For this program can have different result on your console because different compiler can generate different random numbers.

5. Write a C program to determine the value of π by monte-carlo method.

```
#include<bits/stdc++.h>
using namespace std;
int main()
{
    int n,i;
    float r1,r2,m=0,p,ra;
    cout<<"How many numbers you want to generate: ";
    cin>>n;
    for(i=1;i<=n;i++){
        r1=rand()/32768.0;
        r2=rand()/32768.0;

        ra=sqrt(1- pow(r1,2));
        if(r2<=ra)
            m++;
    }
    p=(m/float(n)*4);
    cout<<"\nPi= "<<p<<endl;
    return 0;
}
```

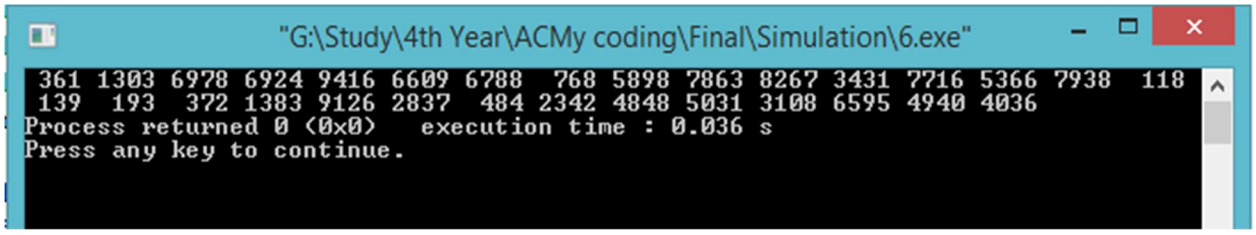
Input & Output:



6. Write a C program for Mid-square Method of Generating 4-digit Random Numbers.

```
#include<stdio.h>
#include<stdlib.h>
#include<math.h>
int main()
{
   long seed=6785,a,b,c,i,j,k,x,y,z;
   for(i=1;i<=30;i++){
      y=seed*seed/100.0;
      z=y/10000.0;
      x=(((y/10000.0)-z)*10000);
      seed=x;
      printf("%4ld ",x);
   }
   return 0;
}
```

Input & Output:

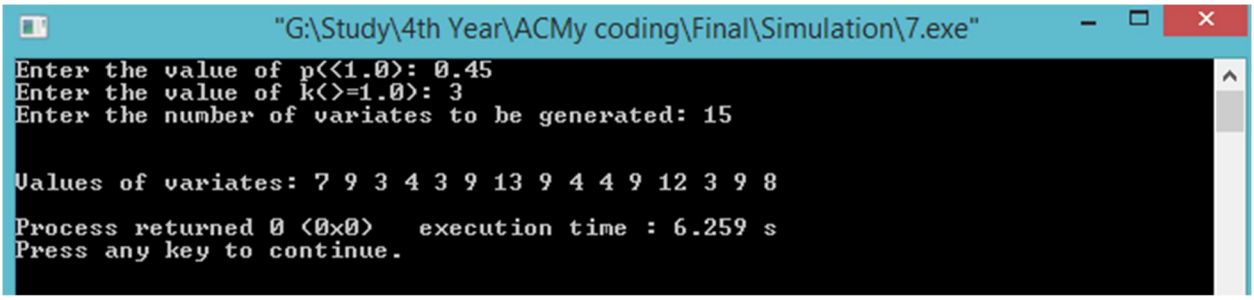


7. Write a C program to generate variables from NEGATIVE BINOMIAL distribution having Parameters k (number of successes) and p (probability of success in trial).

```
#include<bits/stdc++.h>
using namespace std;
int main()
{
   int i,k,s,x,n;
   float p,r;
```

```
    printf("Enter the value of p(<1.0): ");
    scanf("%f",&p);
    printf("Enter the value of k(>=1.0): ");
    scanf("%d",&k);
    printf("Enter the number of variates to be generated: ");
    scanf("%d",&n);
    printf("\n\nValues of variates: ");
    for(i=1;i<=n;i++){
        x=0;
        s=0;
        while(s<k){
            r=rand()/32768.0;
            if(r<=p)
                s++;
            x++;
        }
        printf("%d ",x);
    }
    printf("\n");
    return 0;
}
```

Input & Output:



8. Write a C program to generate random variable from a Normal distribution having mean mue and standard deviation sigma.

```
#include<stdio.h>
#include<stdlib.h>
int main()
{
    int i,j,s,n;
   float meu,sigma,sum,x,t;
   printf("Enter the value of Meu: ");
   scanf("%f",&meu);
   printf("Enter the value of sigma: ");
   scanf("%f",&sigma);
   printf("Enter the number of variates to be generated: ");
   scanf("%d",&n);
   printf("\n\nValues of variates: ");
   for(i=1;i<=n;i++){
      sum=0;
      for(j=0;j<=12;j++){
         x=rand()/32768.0;
         sum+=x;
      }
      t=meu+sigma*(sum-6);
      printf("%.2f ",t);
   }
   printf("\n");
         return 0;
}
```

Input & Output: